CYBERSECURITY PROJECT REPORT

SECURITY OPERATIONS | DETECTION | INVESTIGATION

WINDOWS ENDPOINT SOC MONITORING, DETECTION & INCIDENT INVESTIGATION USING SPLUNK

A Practical Windows Endpoint Security Monitoring Lab

PREPARED BY - **Yalla Vara Prasad**

PROJECT CATEGORY - Security Operations Centre (SOC)

PRIMARY PLATFORM - Splunk Enterprise

INVESTIGATION AREAS -

1. SMB Authentication Failures
2. Encoded PowerShell Process

REPORT VERSION - Version 1.0

HANDS-ON LAB DOCUMENTATION • DETECTION ENGINEERING • SECURITY EVENT ANALYSIS

TABLE OF CONTENTS

S. No.	Section title	Page No.
1	Project Introduction	3
2	Project Objectives	4
3	Lab Architecture and Environment	5 - 6
4	Data Collection and Verification	7 - 8
5	Investigation Report 1 — SMB Authentication Failures	9 - 12
6	Investigation Report 2 — Encoded PowerShell Process	12 - 15
7	Splunk SOC Dashboard	16 - 18
8	Results and Key Findings	19 - 20
9	Challenges and Lessons Learned	20 - 21
10	Conclusion and Future Improvements	22 - 23
11	Appendix A — Screenshot Evidence Index	23 - 24
12	Appendix B — SPL Detection Queries	24 - 26
13	References	27

1. PROJECT INTRODUCTION

Cybersecurity monitoring plays an important role in identifying suspicious activities, investigating security events, and improving the visibility of endpoint systems. Windows endpoints generate different types of security and system logs that can help security analysts understand user activity, authentication failures, and process execution.

This project***, Windows Endpoint SOC Monitoring, Detection & Incident Investigation Using Splunk***, focuses on collecting and analysing Windows endpoint logs through Splunk Enterprise. A Windows 10 virtual machine was configured with Splunk Universal Forwarder and Sysmon to collect Windows event logs and process-creation telemetry. The collected events were forwarded to Splunk Enterprise running on a separate Windows 11 host for searching, analysis, and visualisation

The project includes two controlled security investigations. The first investigates SMB authentication failures and account-lockout activity using Windows Security events. The second examines PowerShell process creation, including a harmless encoded-command simulation, using Sysmon telemetry. Splunk searches were used to identify relevant events, review supporting fields, and develop detection logic for these activities.

An interactive Splunk dashboard was created to present endpoint event volume, authentication activity, process executions, event-source distribution, and selected detection results in one place. The project demonstrates practical skills in Windows log collection, endpoint monitoring, Search Processing Language (SPL), security-event investigation, and dashboard development in a controlled lab environment.

2. PROJECT OBJECTIVES

The main objective of this project is to develop a practical Windows endpoint monitoring and investigation environment using Splunk Enterprise. The project focuses on collecting endpoint telemetry, identifying relevant security events, and presenting investigation findings through searches and dashboards.

The specific objectives are:

1. **Centralised Log Collection:** Configure Splunk Universal Forwarder to collect Windows Security, System, and Application logs from a Windows endpoint and forward them to Splunk Enterprise.

2. **Endpoint Process Monitoring:** Integrate Sysmon telemetry to observe process-creation activity and examine fields such as process image, command line, parent process, and user.

3. **Authentication Failure Investigation:** Analyse SMB authentication failures and account-lockout events using Windows Security Event IDs 4625 and 4740.

4. **PowerShell Activity Detection:** Examine PowerShell process-creation events and identify the use of the -EncodedCommand parameter in a controlled, harmless lab simulation.

5. **SPL-Based Detection:** Develop and use Splunk Search Processing Language (SPL) queries to filter, correlate, summarise, and investigate relevant endpoint events.

6. **Security Dashboard Development:** Build a Splunk dashboard that presents event volume, authentication activity, process executions, event-source distribution, and selected detection results.

7. **Practical SOC Skill Development:** Practise the core workflow of security monitoring, alert investigation, evidence review, and documenting findings in a controlled lab environment.

The project aims to demonstrate how endpoint telemetry and log analysis can support security operations. The lab results represent controlled simulations and should not be interpreted as evidence of an actual compromise.

3. LAB ARCHITECTURE AND ENVIRONMENT

3.1 Lab Overview

The project was implemented in a controlled virtual lab environment using a physical Windows 11 host and a Windows 10 virtual machine. Splunk Enterprise was installed on the Windows 11 host to receive, search, analyse, and visualise endpoint events. The Windows 10 endpoint used Splunk Universal Forwarder to transmit selected Windows event logs and Sysmon telemetry to the central Splunk instance.

A Kali Linux virtual machine was used to generate controlled SMB authentication-failure activity against the lab endpoint. These activities provided sample events for authentication-failure and account-lockout investigations. A separate harmless PowerShell simulation was used to examine process-creation telemetry.

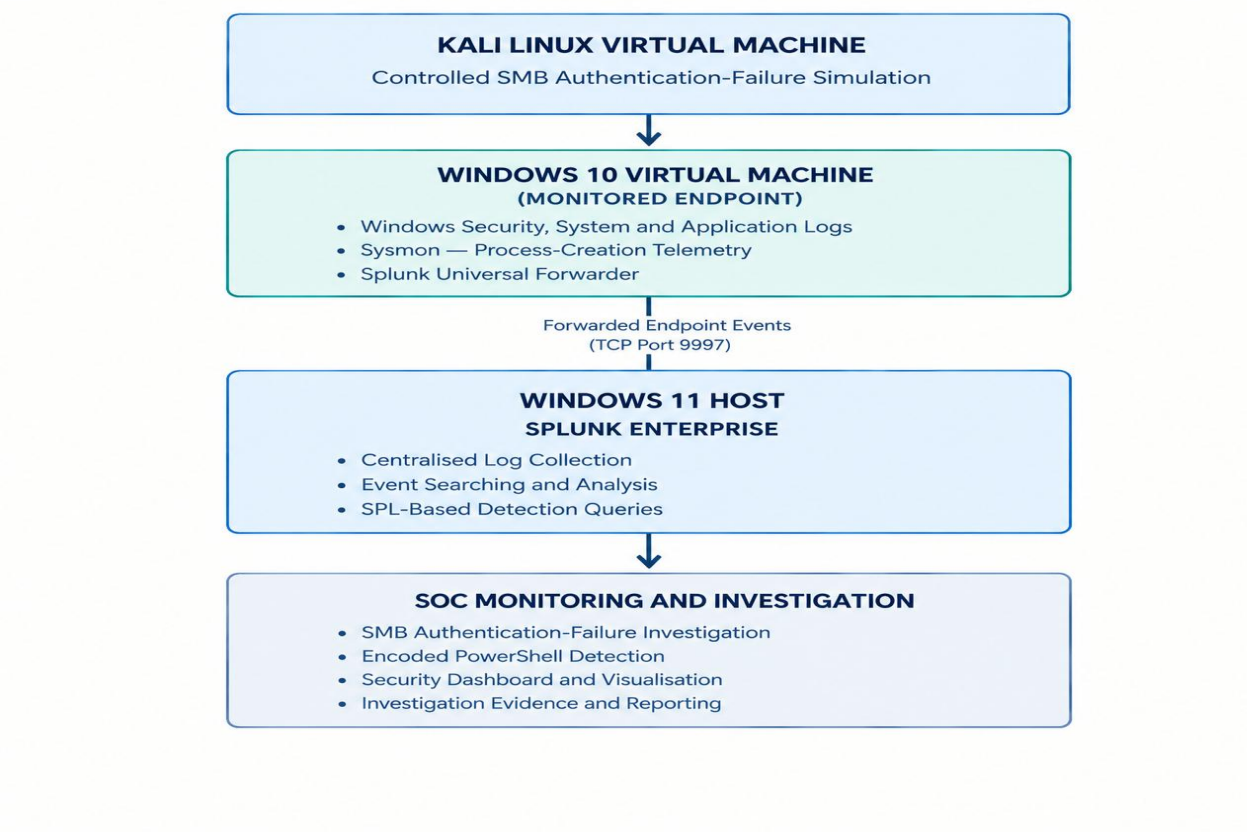


Figure 1. Architecture of the Windows endpoint SOC monitoring and investigation lab, showing the controlled SMB authentication-failure test source, Windows endpoint log collection through Splunk Universal Forwarder and Sysmon, centralised analysis in Splunk Enterprise, and the resulting SOC searches, detections, and dashboard visualisations.

3.2 Lab Components

1. Windows 11 Host — Splunk Enterprise

Hosted the central Splunk Enterprise instance. It received forwarded endpoint data and provided the search interface and SOC dashboard for event analysis.

2. Windows 10 Virtual Machine — Monitored Endpoint

Served as the endpoint under investigation. Windows Security, System, and Application logs were configured for collection through Splunk Universal Forwarder.

3. Splunk Universal Forwarder

Collected the configured Windows event logs from the Windows 10 endpoint and forwarded them to Splunk Enterprise over the configured receiving port.

4. Sysmon

Provided additional endpoint telemetry, including process-creation events. Event details such as the process image, parent image, user, and available command-line information supported the PowerShell investigation.

5. Kali Linux Virtual Machine — Test Source

Was used for controlled SMB authentication-failure testing against the lab endpoint. The resulting Windows Security events were reviewed in Splunk.

3.3 Data Flow

The lab followed this data flow:

Windows 10 endpoint and Sysmon → Splunk Universal Forwarder → Splunk Enterprise on Windows 11 → SPL searches and detection logic → SOC dashboard and investigation evidence.

The lab was designed for authorised testing within the user's own controlled environment. The simulated activities were used to validate log collection and investigation workflows rather than to represent a real-world compromise.

4. DATA COLLECTION AND VERIFICATION

4.1 Windows Event Log Collection

The Windows 10 virtual machine was configured to collect selected Windows Security, System, and Application event logs using Splunk Universal Forwarder. The collected events were forwarded to Splunk Enterprise running on the Windows 11 host. This configuration provided a central location for searching and analysing endpoint activity.

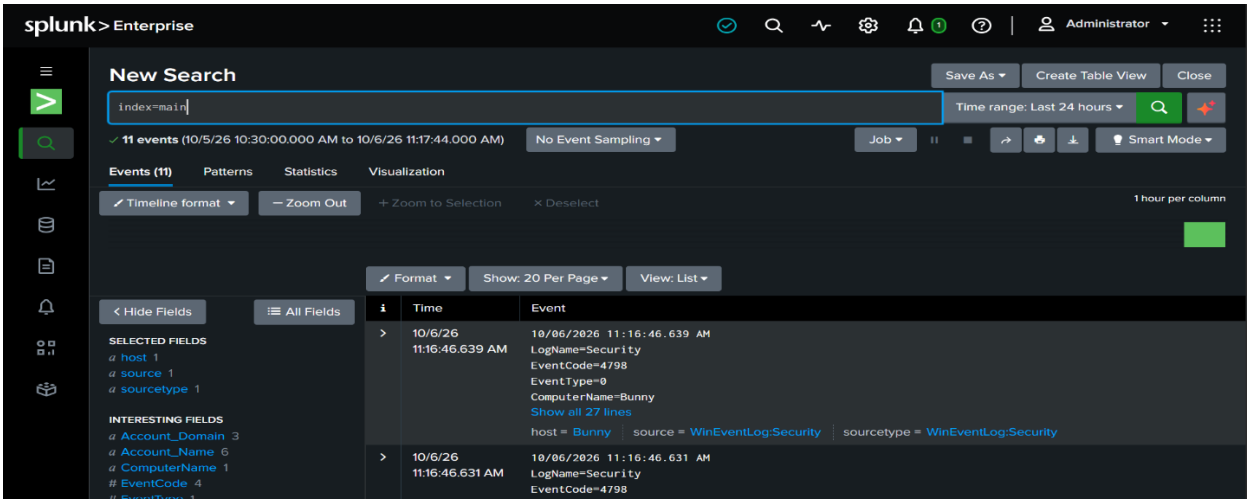


Figure 2. Verification of Windows endpoint event ingestion in Splunk Enterprise using the index=main search query.

4.2 Sysmon Telemetry Collection

Sysmon was configured on the Windows 10 endpoint to provide additional information about system activity, including process-creation events. The Sysmon Operational event channel was collected through Splunk Universal Forwarder. This telemetry supported the investigation of PowerShell process execution by providing fields such as the process image, parent image, user, and available command-line information.

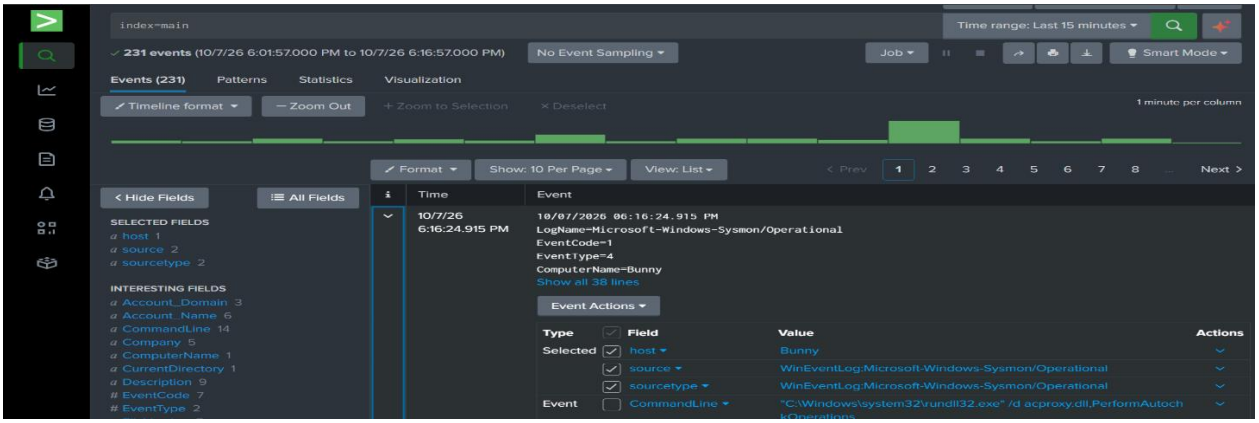


Figure 3. Verification of Sysmon event ingestion in Splunk Enterprise using the index=main search query.

4.3 Forwarding and Search Verification

After configuring the log inputs and forwarding destination, the incoming events were verified in Splunk Enterprise. Searches were used to confirm that endpoint events were searchable in the configured index and to inspect their host, source, sourcetype, and event fields.

The verification process confirmed that the lab could collect and analyse Windows event logs and Sysmon telemetry for the planned investigations. Configuration changes were followed by the required Universal Forwarder service restart where applicable.

4.4 Event Sources Used

The principal data sources used in this project were:

- Windows Security event logs for authentication failures and account-lockout activity.
- Windows System and Application event logs for additional endpoint events.
- Sysmon Operational events for process-creation telemetry.
- Splunk Enterprise searches for event filtering, aggregation, and investigation.

These data sources formed the foundation for the SMB authentication investigation, PowerShell process investigation, and SOC dashboard.

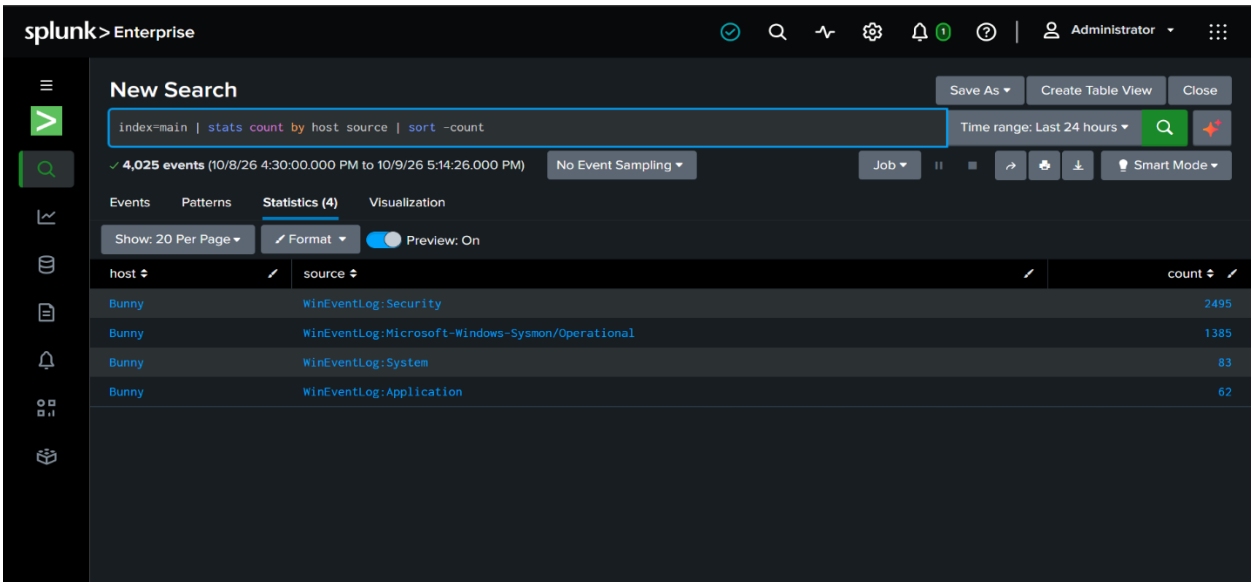


Figure 4. Identification and organisation of Windows endpoint event sources in Splunk Enterprise for log verification and security analysis.

5. INVESTIGATION REPORT 1 — SMB AUTHENTICATION FAILURES AND ACCOUNT LOCKOUT

5.1 Investigation Overview

The first investigation focused on identifying and analysing SMB authentication failures generated in the controlled lab environment. The test was performed using a Kali Linux virtual machine to simulate unsuccessful authentication attempts against the Windows 10 endpoint. The resulting Windows Security events were collected by Splunk Universal Forwarder and analysed in Splunk Enterprise.

The objective was to verify that failed authentication activity could be detected, investigate the associated event details, and identify account-lockout activity where recorded.

5.2 Test Procedure

A controlled SMB authentication test was performed against the lab endpoint using the test account SOC_LabTest and the lab share SOC_LabShare. Unsuccessful authentication attempts generated Windows Security events. During the test, the account subsequently entered a locked-out state, and further attempts were stopped once the lockout was observed.

The test was limited to the authorised lab environment. It was conducted to validate log collection and investigation methods, not to access an external system or demonstrate a real-world compromise.

5.3 Security Events Observed

Two Windows Security event IDs were important to this investigation:

- **Event ID 4625 — Failed Logon:** Records an unsuccessful logon attempt. Depending on the event, relevant details can include the account, logon type, failure reason, and source network address.

- **Event ID 4740 — Account Locked Out:** Records an account-lockout event and can provide information about the affected account and caller computer.

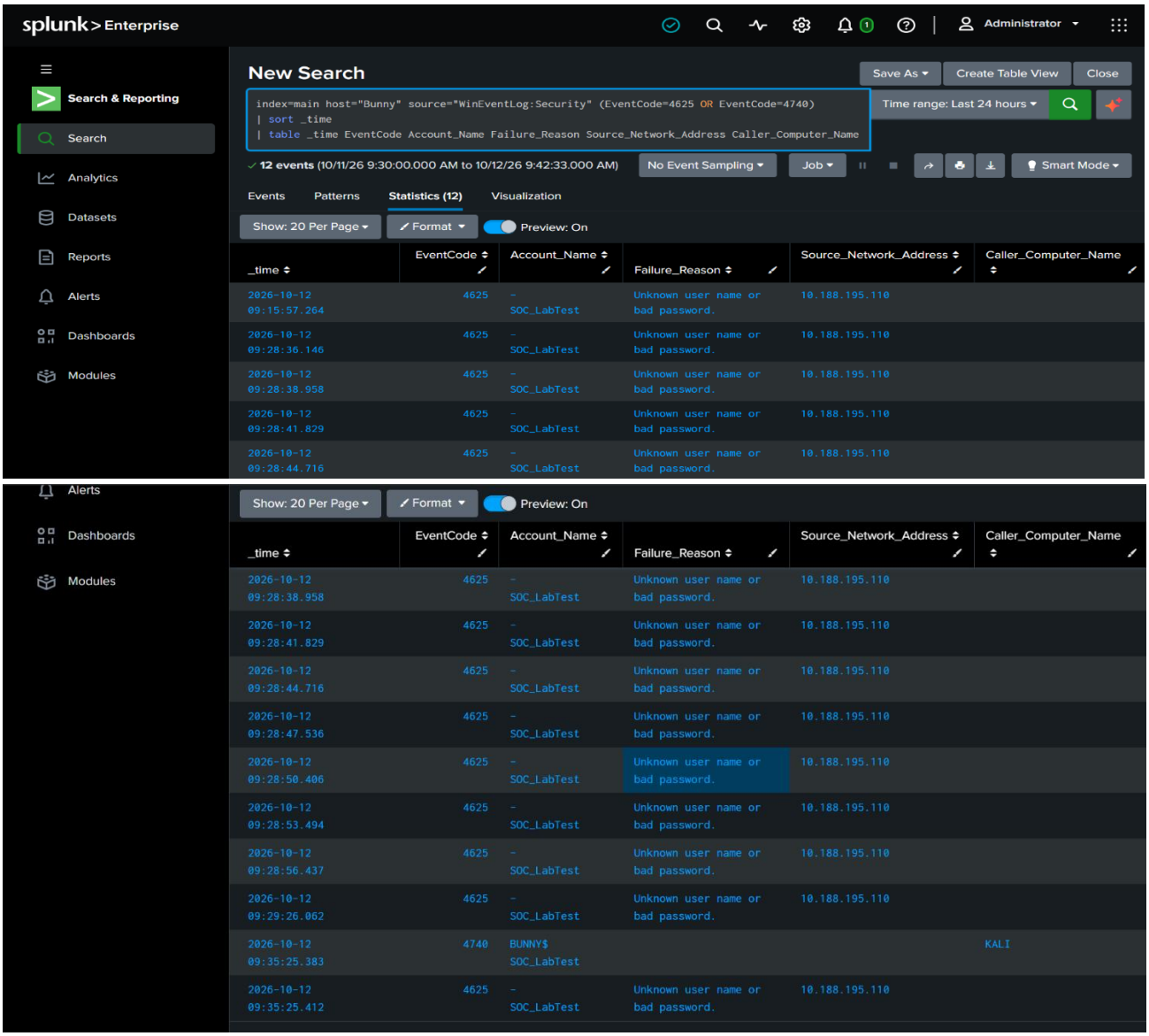


Figure 5(a) & Figure 5(b)

These events were reviewed in Windows Event Viewer and searched in Splunk Enterprise. Their presence helped verify that the test activity was recorded by the endpoint and forwarded for central analysis.

5.4 Splunk Investigation Method

Splunk searches were used to filter Windows Security events, review authentication-related fields, and summarise failed logons. A time-based aggregation was also used to identify periods in which the number of failed logons reached the configured threshold.

A separate search was used to group failed logons by account and source address. A threshold search helped highlight accounts with repeated failures within a one-minute interval. The results were reviewed in context rather than treating a high failure count alone as proof of malicious activity.

SPL 1 — Search for failed logons and account lockouts

```
index=main host="Bunny" source="WinEventLog:Security"
(EventCode=4625 OR EventCode=4740)
| sort 0 _time
| table _time EventCode Account_Name Logon_Type Failure_Reason Source_Network_Address Caller_Computer_Name
```

SPL 2 — Summarise failed logons by account and source

```
index=main host="Bunny" source="WinEventLog:Security" EventCode=4625
| stats count as FailedLogons by Account_Name Source_Network_Address Logon_Type Failure_Reason
| sort - FailedLogons
```

SPL 3 — Display failed-logon details for investigation

```
index=main host="Bunny" source="WinEventLog:Security" EventCode=4625
| where isnotnull(Account_Name) AND Account_Name!="" AND Account_Name!="-"
| table _time Account_Name Source_Network_Address Logon_Type Failure_Reason
| sort 0 - _time
| head 10
```

SPL 4 — Detect repeated failures within one minute

```
index=main host="Bunny" source="WinEventLog:Security" EventCode=4625
| where isnotnull(Account_Name) AND Account_Name!="" AND Account_Name!="-"
| bin _time span=1m
| stats count as FailedLogons by _time Account_Name Source_Network_Address
| where FailedLogons >= 5
| sort 0 - FailedLogons
| head 10
```

5.5 Findings

The controlled test produced failed-authentication activity, and an account-lockout event was observed. Splunk provided a central interface for examining the relevant Windows Security events and reviewing the available account and logon details.

The investigation demonstrated how an analyst can use event IDs, timestamps, account information, and source details to investigate repeated authentication failures. The observed activity was generated deliberately as part of the lab, so it should be interpreted as a successful detection and log-analysis test rather than evidence of an actual attack.

5.6 Conclusion

This investigation validated the collection and analysis of SMB authentication-failure and account-lockout events in Splunk. It provided practical experience in filtering Windows Security logs, reviewing authentication fields, and applying threshold-based searches to identify repeated failures. The workflow illustrates how endpoint event data can support an initial SOC investigation and help analysts decide when further review is required.

6. INVESTIGATION REPORT 2 — ENCODED POWERSHELL PROCESS DETECTION AND INVESTIGATION

6.1 Investigation Overview

The second investigation focused on monitoring PowerShell process creation on the Windows 10 endpoint. Sysmon telemetry was collected through Splunk Universal Forwarder and analysed in Splunk Enterprise to identify process activity and examine relevant process details.

A harmless encoded-command simulation was performed to understand how the -EncodedCommand parameter appears in process-creation telemetry. This investigation demonstrated how process information can be searched and reviewed as part of an endpoint security monitoring workflow.

6.2 Test Procedure

A controlled PowerShell test was performed on the Windows 10 lab endpoint. A harmless text-output command was encoded and executed using PowerShell's -EncodedCommand parameter. The test was designed to generate process-creation telemetry for investigation without performing a malicious action.

Sysmon Event ID 1 was used to examine process-creation activity. The available event fields were reviewed to understand the process image, parent process, user, process identifier, and command-line information where present.

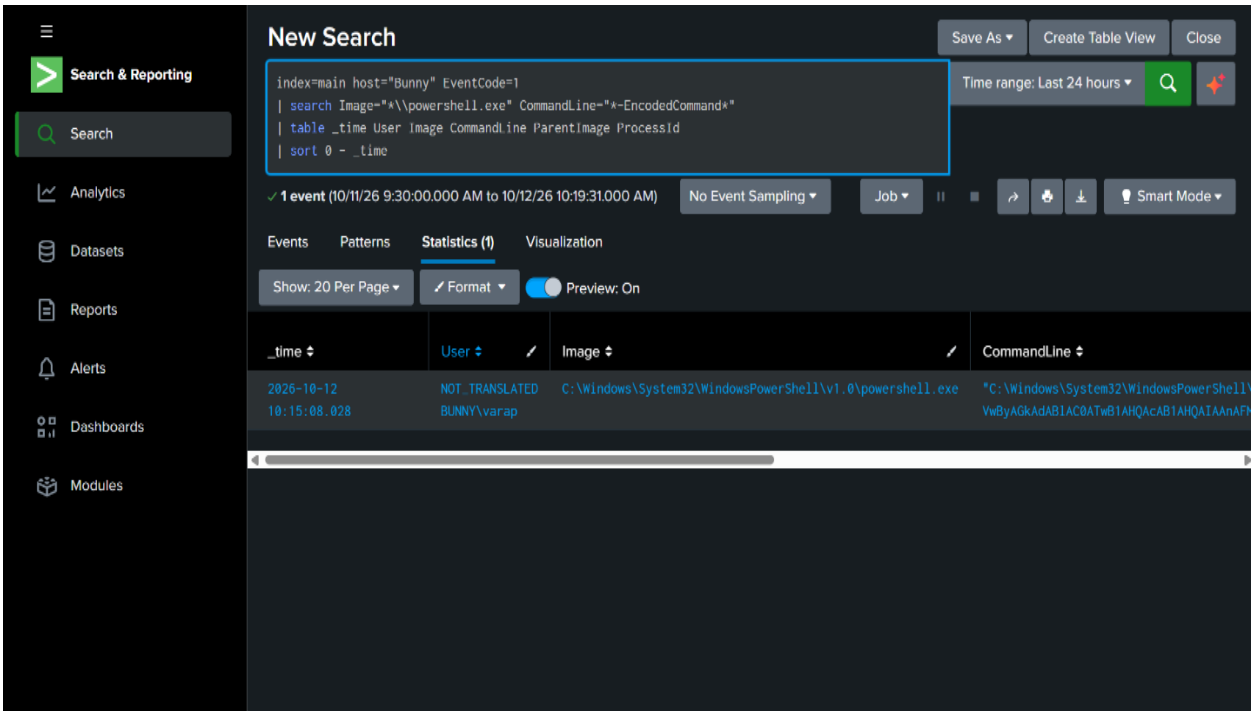
```
PS C:\Users\varap> Write-Output "SOC_SIMULATION_2_START"
SOC_SIMULATION_2_START
PS C:\Users\varap> $payload = "Write-Output 'SOC_LAB_ENCODED_TEST'"
>> $bytes = [System.Text.Encoding]::Unicode.GetBytes($payload)
>> $encoded = [Convert]::ToBase64String($bytes)
>> powershell.exe -NoProfile -EncodedCommand $encoded
SOC_LAB_ENCODED_TEST
PS C:\Users\varap>
```

Figure 6. Sysmon process-creation event identifying PowerShell execution with the -EncodedCommand parameter during the controlled lab simulation.

6.3 Sysmon Event Analysis

Sysmon Event ID 1 records process-creation activity. In this investigation, the event data was used to identify PowerShell execution and inspect the context surrounding the process.

The -EncodedCommand parameter was treated as an indicator requiring review, not as proof of malicious behaviour. Encoded commands can be used for legitimate administrative tasks as well as in suspicious activity, so process context and the command's purpose must be considered.



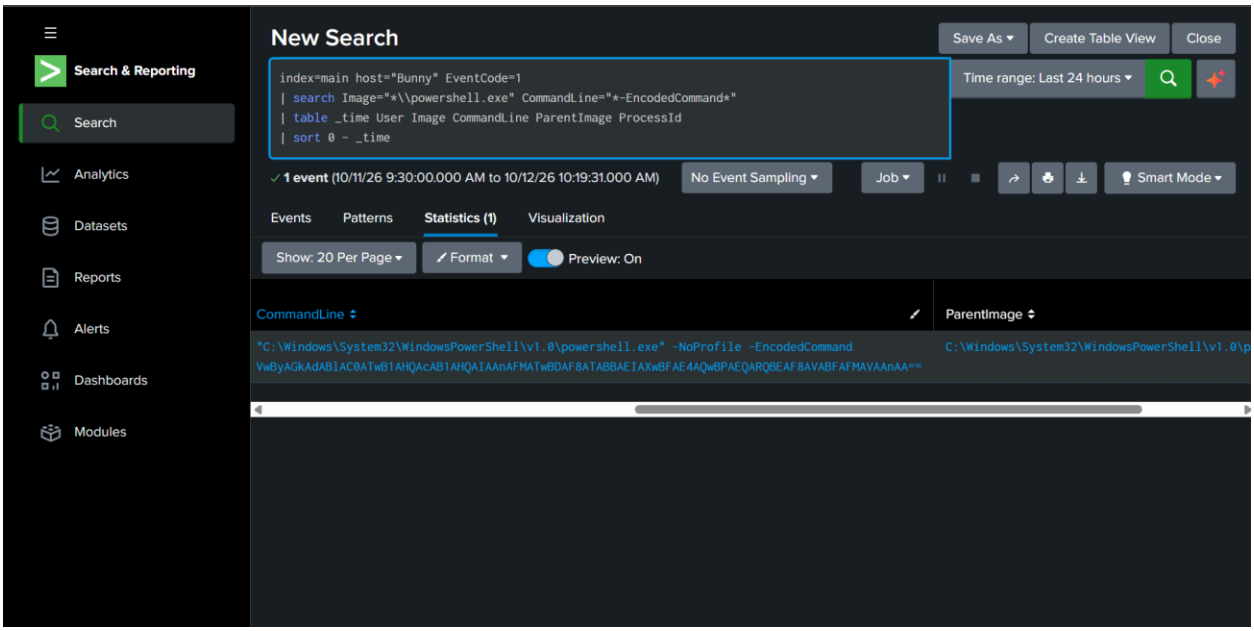


Figure 7(a) & Figure 7(b). Sysmon process-creation event identifying PowerShell execution with the -EncodedCommand parameter during the controlled lab simulation.

6.4 Splunk Investigation Method

Splunk was used to filter Sysmon process-creation events and identify PowerShell executions with the encoded-command parameter in the available command-line field.

The query filters for PowerShell process-creation events and displays selected investigation fields. If the command-line field is missing or does not contain the parameter in a particular event, that event may not match this search.

SPL 1 — Encoded PowerShell Detection

```
index=main host="Bunny" EventCode=1
| search Image="*\\powershell.exe" CommandLine="*-EncodedCommand*"
| table _time User ParentImage ProcessId
| sort 0 - _time
| head 10
```

SPL 2 — Review PowerShell process-creation events

```
index=main host="Bunny" EventCode=1
| search Image="*\\powershell.exe"
| table _time User Image CommandLine ParentImage ProcessId
| sort 0 - _time
| head 10
```

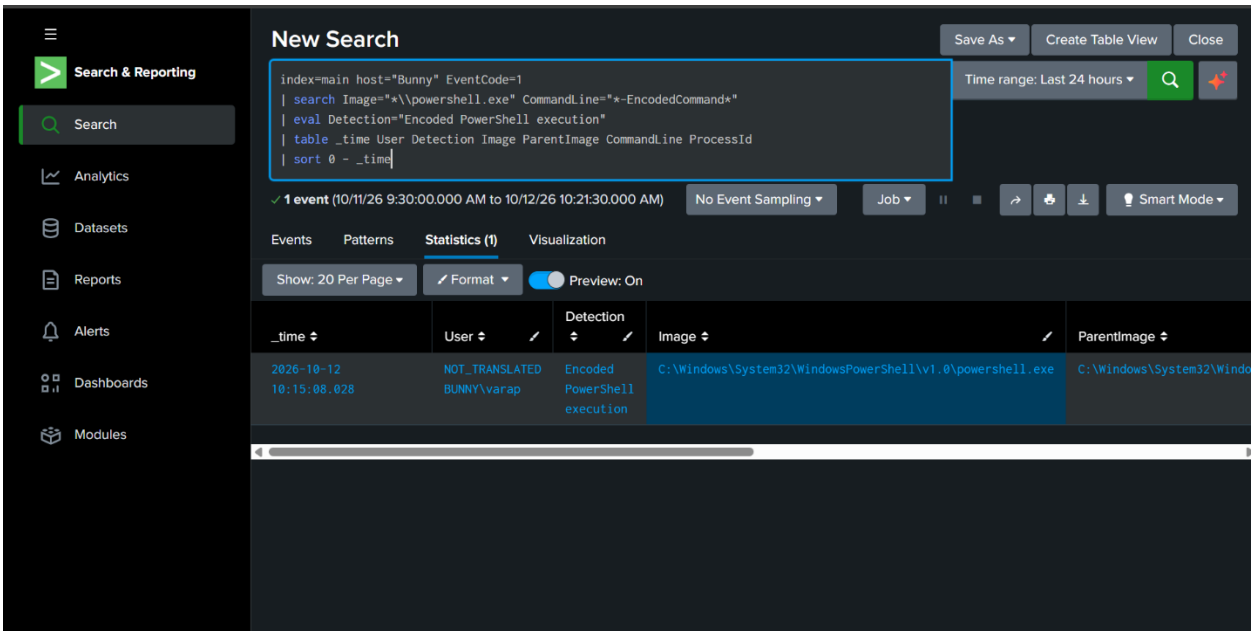


Figure 8. Splunk detection results identifying PowerShell process-creation events containing the -EncodedCommand parameter during the controlled lab simulation.

6.5 Findings

The investigation used Sysmon process-creation telemetry to examine PowerShell activity on the Windows endpoint. The controlled encoded-command test provided a safe example for understanding how PowerShell execution can be investigated through process metadata and Splunk searches.

The investigation demonstrated the value of reviewing the process image, parent process, user, process identifier, and command-line information when available. The encoded-command parameter was treated as a review indicator, and the lab simulation was not considered evidence of an actual compromise.

6.6 Conclusion

This investigation provided practical experience in collecting and searching Sysmon process-creation events, filtering PowerShell activity, and reviewing endpoint process metadata. It demonstrated how SPL searches can help an analyst identify activity for further investigation while avoiding the assumption that an individual command-line indicator automatically represents malicious behaviour.

7. SPLUNK SOC DASHBOARD

7.1 Dashboard Overview

A Splunk dashboard titled **Windows Endpoint SOC Operations** was developed to provide a centralised view of Windows endpoint activity and selected security detections. The dashboard brings together event-volume summaries, authentication activity, process executions, event-source distribution, and investigation results in a single interface.

7.2 Dashboard Components

The Windows Endpoint SOC Operations dashboard contains the following components:

1. **Total Events:** Displays the overall number of events collected by Splunk from the monitored endpoint.
2. **Failed Logons:** Displays the number of failed authentication events identified in the selected data and time range.
3. **Event Source Breakdown:** Shows how the collected events are distributed across different event sources.
4. **Endpoint Activity Timeline:** Presents endpoint event activity over time to help identify changes or unusual periods of activity.
5. **Process Execution Monitor:** Displays process-creation activity collected through Sysmon for endpoint monitoring and investigation.
6. **Sysmon EventCode Distribution:** Summarises Sysmon events by event ID to provide an overview of the types of endpoint activity recorded.
7. **Authentication Activity:** Presents authentication-related events for reviewing failed logons and related account activity.
8. **Authentication Investigation Feed:** Provides event-level details that support investigation of authentication activity.
9. **Brute-Force Authentication Detection:** Highlights repeated failed-logon activity that meets the configured detection threshold.
10. **PowerShell Detection Feed:** Displays PowerShell process-creation events matching the encoded-command detection search.
11. **Failed Logons Over Time:** Visualises failed authentication activity over time to help identify periods with repeated failures.

These components combine event summaries, visualisations, and detailed investigation results to support endpoint monitoring and security-event analysis.

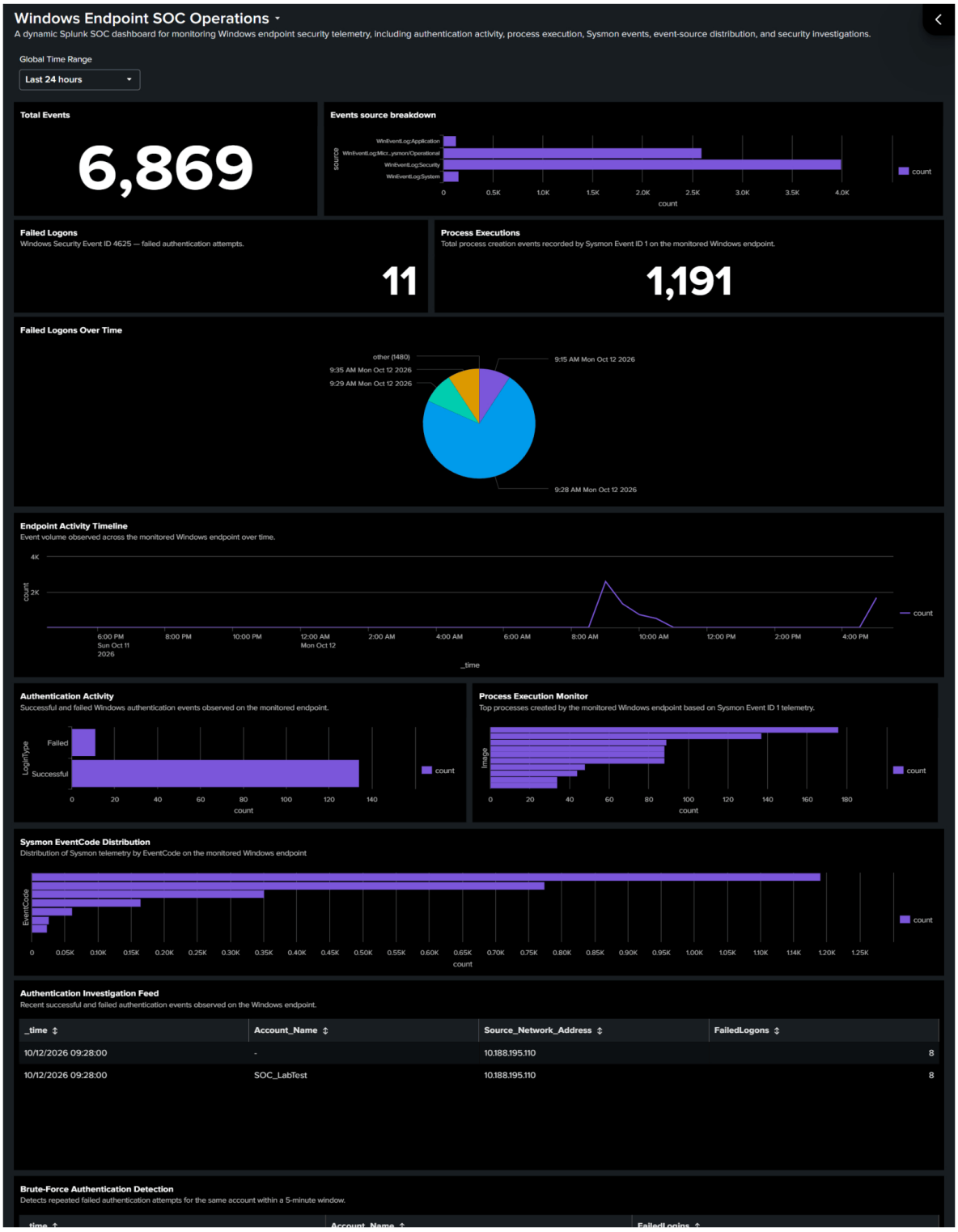


Figure 9. Windows Endpoint SOC Operations dashboard in Splunk Enterprise, presenting endpoint event volume, authentication activity, process execution monitoring, event-source distribution, and selected security detection results.

7.3 Dashboard Results

The dashboard provides the following outcomes:

- **Centralised visibility:** Brings multiple Windows endpoint event sources together in one interface.
- **Authentication monitoring:** Makes failed-logon activity easier to review.
- **Process monitoring:** Provides visibility into process-creation events collected through Sysmon.
- **Detection support:** Presents results from the configured repeated-failure and encoded-PowerShell searches.
- **Time-based analysis:** Helps show changes in endpoint activity and failed logons over time.
- **Investigation support:** Combines summary panels with detailed event information for further analysis.

The dashboard demonstrates how Splunk searches and visualisations can support a structured SOC monitoring workflow. Its findings reflect the configured lab data and controlled simulations, not a confirmed real-world compromise.

8. RESULTS AND KEY FINDINGS

The project was evaluated through controlled lab simulations, endpoint event collection, Splunk searches, and dashboard visualisation. The key results are summarised below.

8.1 Key Results

8.1. 1 Windows Event Monitoring and Authentication Results

The Splunk dashboard provides an overview of collected Windows endpoint events and failed-logon activity over time. These visualisations help monitor endpoint activity and identify periods of increased authentication failures for further investigation.



Figure 10. Splunk dashboard showing total endpoint events and failed-logon activity over time.

8.1.2 Encoded PowerShell Detection Results

Sysmon process-creation events were analysed in Splunk to identify PowerShell executions containing the -EncodedCommand parameter. The search results demonstrate how process telemetry and SPL queries can help identify activity requiring further investigation. Encoded commands alone do not confirm malicious behaviour.



Figure 11. Splunk search results for PowerShell process-creation events containing the -EncodedCommand parameter**.**

8.2 Overall Outcome

The project demonstrated an end-to-end endpoint monitoring and investigation workflow, from collecting Windows event data to reviewing security events and presenting selected results through a Splunk dashboard. The two controlled investigations provided practical experience in authentication-event analysis, process monitoring, and threshold-based detection.

The results validate the configured lab workflow for the tested scenarios. They do not establish that a real-world attack occurred or that the detection searches identify every possible malicious activity.

9. CHALLENGES AND LESSONS LEARNED

During the implementation of this project, several configuration and investigation tasks provided practical learning opportunities.

9.1 Windows Event-Log Collection

- **Challenge:** Collecting Windows Security, System, and Application logs in Splunk.
- **Action Taken:** Configured Splunk Universal Forwarder to collect and forward the required event logs.
- **Lesson Learned:** Correct input and output configurations are essential for centralised log collection.

9.2 Network IP Address Changes

- **Challenge:** The Windows 11 host IP address changed when using the mobile hotspot.
- **Action Taken:** Updated the forwarding destination and verified that endpoint events were arriving in Splunk.
- **Lesson Learned:** Network changes can interrupt log forwarding, so connectivity and configuration must be checked.

9.3 Sysmon Event Collection

- **Challenge:** Configuring access to Sysmon Operational events.
- **Action Taken:** Verified the event channel and addressed the permissions required for collection.
- **Lesson Learned:** Correct permissions and event-channel configuration are necessary for endpoint monitoring.

9.4 Event-Field Analysis

- **Challenge:** Understanding the fields available in Windows Security and Sysmon events.
- **Action Taken:** Examined event IDs, account names, source addresses, process images, and parent-process details.
- **Lesson Learned:** Understanding event fields helps analysts identify relevant evidence during investigations.

9.5 SMB Authentication Investigation

- **Challenge:** Identifying repeated failed authentication attempts and account-lockout activity.
- **Action Taken:** Investigated Windows Security Event IDs 4625 and 4740 using Splunk searches.
- **Lesson Learned:** Reviewing related security events provides better context for authentication investigations.

9.6 PowerShell Process Detection

- **Challenge:** Identifying PowerShell process activity associated with an encoded command.
- **Action Taken:** Executed a harmless encoded-command simulation and analysed the resulting Sysmon process-creation telemetry.
- **Lesson Learned:** Detection indicators require contextual investigation and do not automatically prove malicious behaviour.

9.7 SPL Query Development

- **Challenge:** Filtering and summarising relevant endpoint events.
- **Action Taken:** Developed SPL searches to investigate authentication failures, apply thresholds, and examine PowerShell process activity.
- **Lesson Learned:** Accurate filters, time intervals, and field selection are important for meaningful detection results.

9.8 Dashboard Development

- **Challenge:** Organising multiple visualisations and investigation panels into a readable dashboard.
- **Action Taken:** Developed the Windows Endpoint SOC Operations dashboard in Splunk Dashboard Studio.

.

10. CONCLUSION AND FUTURE IMPROVEMENTS

10.1 Conclusion

The project, ***Windows Endpoint SOC Monitoring, Detection & Incident Investigation Using Splunk,*** demonstrated a practical workflow for collecting, monitoring, and investigating Windows endpoint events.

The key outcomes were:

- **Centralised log collection:** Configured Splunk Universal Forwarder to send Windows event logs to Splunk Enterprise.
- **Endpoint monitoring:** Collected and analysed Sysmon process-creation telemetry.
- **Authentication investigation:** Examined SMB authentication failures and account-lockout events using Windows Security Event IDs 4625 and 4740.
- **PowerShell investigation:** Used a harmless encoded-command simulation to examine PowerShell process-creation activity.
- **SPL-based detection:** Applied Splunk searches to filter, summarise, and investigate relevant security events.
- **Dashboard visualisation:** Developed the Windows Endpoint SOC Operations dashboard to present event summaries, authentication activity, process monitoring, and selected detection results.

Overall, the project provided practical experience in endpoint telemetry collection, security-event analysis, detection logic, and dashboard development. The investigations were performed in a controlled lab environment and do not represent a confirmed real-world security incident.

10.2 Future Improvements

The project can be extended in the following ways:

1. **Additional Detection Rules:** Develop more detections for suspicious logon patterns, privilege changes, and unusual process execution.
2. **Splunk Alerting:** Configure alerts for repeated authentication failures and selected suspicious process activity.
3. **Improved Detection Context:** Enrich events with additional fields and contextual information to support more detailed investigations.

4. **MITRE ATT&CK Mapping:** Map relevant detections to applicable MITRE ATT&CK techniques to improve investigation documentation.
5. **Expanded Endpoint Coverage:** Add more authorised Windows endpoints to evaluate centralised monitoring across multiple systems.
6. **Dashboard Enhancements:** Improve dashboard filtering, time-range selection, drill-down searches, and panel organisation.
7. **Detection Testing:** Evaluate detection rules using additional controlled scenarios and document false positives and detection limitations.

These improvements would help extend the lab into a broader endpoint monitoring and detection environment while providing further opportunities to practise SOC investigation workflows.

11. APPENDIX A — SCREENSHOT EVIDENCE INDEX

The following figures document the configuration, testing, investigation, and dashboard results of the Windows Endpoint SOC Monitoring project.

Figure 1 : Architecture of the Windows endpoint SOC monitoring and investigation lab.

Figure 2 : Verification of Windows endpoint event ingestion in Splunk Enterprise using the index=main search query.

Figure 3 : Verification of Sysmon event ingestion in Splunk Enterprise using the index=main search query.

Figure 4 : Identification and organisation of Windows endpoint event sources in Splunk Enterprise.

Figure 5(a & b) : Windows Security events showing SMB authentication failures and account-lockout activity during the controlled lab simulation.

Figure 6 : Sysmon process-creation event showing PowerShell execution with the -EncodedCommand parameter during the controlled lab simulation.

Figure 7(a) : Sysmon process-creation evidence showing PowerShell execution during the controlled encoded-command simulation.

Figure 7(b) : Additional Sysmon process-creation evidence showing PowerShell execution during the controlled encoded-command simulation.

Figure 8 : Splunk detection results identifying PowerShell process-creation events containing the -EncodedCommand parameter during the controlled lab simulation.

Figure 9 : Windows Endpoint SOC Operations dashboard in Splunk Enterprise, presenting endpoint event volume, authentication activity, process execution monitoring, event-source distribution, and selected security detection results.

Figure 10. Splunk dashboard displaying total Windows endpoint events and failed-logon activity over time, supporting endpoint monitoring and authentication-failure analysis.

Figure 11. Splunk search results identifying PowerShell process-creation events containing the -EncodedCommand parameter during the controlled lab investigation.

Only figures included in the final report should appear in this index. Figure numbers and descriptions should match the screenshots and captions used in the report.

12. APPENDIX B — SPL DETECTION QUERIES

12.1 Purpose

This appendix contains selected Splunk Search Processing Language (SPL) queries used during the Windows endpoint SOC monitoring project. The queries support Windows event review, SMB authentication-failure investigation, repeated failed-logon detection, and encoded PowerShell process monitoring.

The searches were executed against endpoint telemetry collected in the controlled lab environment. Results depend on the available events, field extraction, host configuration, and selected time range.

12.2 Windows Endpoint Event Review

The following query was used to review collected Windows endpoint events and inspect their source information.

```
index=main host="Bunny"
| table _time host source sourcetype
| head 20
```

Purpose: Verify searchable endpoint events and review their timestamps, host, source, and sourcetype.

12.3 SMB Failed-Logon Investigation

This query filters Windows Security Event ID 4625 and displays available fields relevant to failed-logon analysis.

```
index=main host="Bunny" source="WinEventLog:Security" EventCode=4625
| table _time Account_Name Source_Network_Address Logon_Type Failure_Reason
```

```
| sort 0 - _time
| head 10
```

Purpose: Review recent failed-logon events and examine account, source address, logon type, and failure reason where available.

12.4 Failed-Logon Aggregation

This query groups failed-logon events by account and source address.

```
index=main host="Bunny" source="WinEventLog:Security" EventCode=4625
| stats count as FailedLogons by Account_Name Source_Network_Address Logon_Type Failure_Reason
| sort - FailedLogons
```

Purpose: Summarise failed-logon activity to help identify accounts or sources associated with repeated failures.

12.5 Repeated Failed-Logon Threshold

This query groups failed-logon events into one-minute intervals and displays groups with at least five failures.

```
index=main host="Bunny" source="WinEventLog:Security" EventCode=4625
| bin _time span=1m
| stats count as FailedLogons by _time Account_Name
| where FailedLogons >= 5
| sort 0 - FailedLogons
```

Purpose: Identify account and time intervals that meet the configured threshold. A threshold match is an investigation lead, not proof of a brute-force attack.

12.6 Account-Lockout Event Review

This query displays failed-logon and account-lockout events together for investigation.

```
index=main host="Bunny" source="WinEventLog:Security"
(EventCode=4625 OR EventCode=4740)
| sort 0 _time
| table _time EventCode Account_Name Logon_Type Failure_Reason Source_Network_Address
Caller_Computer_Name
```

Purpose: Review failed-logon events (4625) alongside account-lockout events (4740) to understand the sequence of activity where the available fields permit.

12.7 Encoded PowerShell Detection

This query searches Sysmon process-creation events for PowerShell command lines containing the -EncodedCommand parameter.

```
index=main host="Bunny" EventCode=1
| search Image="*\\powershell.exe" CommandLine="*-EncodedCommand*"
| table _time User ParentImage ProcessId Image CommandLine
| sort 0 - _time
| head 10
```

Purpose: Identify PowerShell process-creation events that include the encoded-command parameter for further review. Its presence alone does not establish malicious activity.

12.8 Query Usage Notes

- Select the appropriate time range in Splunk before running each query.
- Confirm that the host, source, event code, and fields match the collected data in the lab.
- Review the full event details before reaching an investigation conclusion.
- Treat threshold results as leads for investigation rather than automatic proof of an attack.
- Redact private IP addresses, usernames, or other sensitive lab details before publishing screenshots or queries publicly.
- These queries document the lab searches and should be adjusted if the field names or event sources differ in another environment.

13. REFERENCES

1. Splunk, “Welcome to the Search Reference,” *Splunk Enterprise Documentation*. https://help.splunk.com/en/splunk-enterprise/spl-search-reference/9.0/introduction/welcome-to-the-search-reference
2. Splunk, “The Universal Forwarder,” *Splunk Documentation*. Available: https://help.splunk.com/en/splunk-cloud-platform/forward-and-process-data/ingest-actions/process-data-with-forwarders/the-universal-forwarder
3. Microsoft, “Sysmon,” *Microsoft Learn — Sysinternals*. Available: https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon
4. Microsoft, “Audit Security System Extension,” *Microsoft Learn*. Available: https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/audit-security-system-extension
5. Microsoft, “Audit User Account Management,” *Microsoft Learn*. Available: https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/audit-user-account-management
6. MITRE, “Command and Scripting Interpreter: PowerShell (T1059.001),” *MITRE ATT&CK*. Available: https://attack.mitre.org/techniques/T1059/001/
7. Splunk, “Install a Windows Universal Forwarder,” *Splunk Documentation*. Available: https://help.splunk.com/en/splunk-cloud-platform/forward-and-process-data/universal-forwarder-manual/9.0/install-the-universal-forwarder/install-a-windows-universal-forwarder
8. Microsoft, “Windows Security Auditing,” *Microsoft Learn*. Available: https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/basic-audit-policy-recommendations

9. Splunk, “Configure an Intermediate Forwarder,” *Splunk Documentation*. Available: https://docs.splunk.com/Documentation/Forwarder/latest/Forwarder/Configureanintermediateforwarder

10. Microsoft, “Windows Event Log,” *Microsoft Learn*. Available: https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log